

The Embedded-Chain RGA

A tombstone-free replicated list whose sort keys are the birth chains,
entropy-coded: design, proofs, validation, and related work

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working names in the artifact: `embed-tree` (unary), `embed-code` (delta-coded)

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Abstract

The embedded-chain RGA (replicated growable array) is a replicated list that keeps no tombstones, no ledger, and no event log: its entire state is the live nodes. It is a sequence MRDT, a *mergeable replicated data type*: state-based, reconciled by a three-way merge against the branches' lowest common ancestor version. A deleted element survives only as $\Theta(\log \delta)$ bits inside its descendants' sort keys, δ being the Lamport-timestamp gap between the element and its own insertion anchor (1 under continuation typing); that is the log of the event gap the insertion crossed, a race or a cursor jump, paid once, in space. Concurrent inserts provably cannot tie, so the merge contains no repair and never decides an order; the state is a canonical function of the event set, and convergence is literal equality. Observationally the datatype *is* the published RGA with the tombstone set compressed into coordinate entropy: a kernel-checked read-equivalence theorem, with machine lockstep at every step as its tested form. All of this follows from one idea: an element's sort key is its *birth chain*, the timestamps from the root of the birth tree down to the element, dead ancestors included. The chain is written at birth, never edited, and carried in an order-preserving self-delimiting code. Separating the datatype from its encoding makes the metatheory nearly definitional: correctness is three one-line facts about chains, plus a single faithfulness theorem about the code. The design is machine-validated (the L1–L25 anomaly battery, except L19, RGA's own, discharged by the sided generalization of §5; 420+ randomized version-DAG executions; 120/120 lockstep), and the central theorem, RA-linearizability at every honestly reachable configuration, is a kernel-checked Lean theorem (§8).

1 The datatype: sort keys are birth chains

Timestamps $t \in \mathbb{N}^+$ are Lamport stamps, globally unique, and double as element identities. Causality gives: the anchor of every insert carries a smaller timestamp than the insert itself.

The birth tree. Every insert names an anchor. The event set therefore induces a tree: the root is the front sentinel 0; the *birth parent* of each element is its anchor. The tree is append-only and never edited: an element's place in it is fixed at birth.

Birth chains. The *birth chain* of u is the sequence of timestamps from the root down to u (sentinel elided): $\text{chain}(u) = \text{chain}(a) \cdot u$ when u was inserted after anchor a , and $\text{chain}(u) = \langle u \rangle$ at the front. Dead ancestors are *not* elided: their timestamps remain in their descendants' chains. The

chain is the sort key, and everything in this section is stated in terms of it; nothing here depends on how chains are stored (§2).

Type. A state is a finite map over *live nodes only*:

$$\sigma : u \mapsto (\text{chain}(u), \text{char}_u).$$

init. $\sigma_0 = \emptyset$.

do (the operations). Two operations, with $\text{rc} = \text{Either}$ everywhere (§3):

- $\text{ins}(x \leftarrow a, c, \pi)$: insert element x (its timestamp) with character c after anchor a ($a = 0$ for the front). The payload $\pi = \text{chain}(a)$ is the anchor’s birth chain, carried *for the proof alone* (§3). Application on a state where a is live (every reachable application, by honesty):

$$\sigma[x] := (\text{chain}_\sigma(a) \cdot x, c), \quad \pi \text{ unread.}$$

No head-tracking, no reading of siblings.

- $\text{del}(d)$: remove d ’s record. Nothing else changes: survivors’ chains are untouched; in particular, d ’s descendants still carry d ’s timestamp in their sort keys. If d is absent, no-op (concurrent double delete).

read. Sort survivors by *chain-lex*: compare chains timestamp-by-timestamp with *larger first*; a proper prefix precedes its extensions (a live ancestor displays immediately before its descendants). Among same-parent siblings this is exactly RGA’s newest-first rule; equivalently, the canonical order is the RGA traversal of the birth tree restricted to survivors.

merge. $\text{merge}(L, A, B)$, where L is the LCA (the branches’ lowest common ancestor version, the merge context the MRDT model stores and supplies). The merge is *survival* (OR-set): live iff in all three, or born in exactly one branch. There is no second step at this layer: survivors’ chains come with their insert events, so every input that holds a survivor agrees on its chain. The merge never compares timestamps and never decides an order.

Inv / applicable / honesty. The honesty contract is the standard RGA generation discipline: an insert is *generated* at a replica where its anchor is live (you insert after a character you can see), and ins/del are *applied* on states where they are applicable (anchor live; delete of a live or already-dead node). On every reachable state, application never consults π . Canonicity (Theorem 1) doubles as Inv; at the representation layer, Theorem 2(iii) is its stored form.

$\approx$. Equality. The state is a canonical function of the event set (Theorem 1), so no quotient is needed: convergence is literal state equality.

The metatheory, which is nearly definitional. At this layer the design’s theorems all but evaporate; we record them because they are the specification the representation must meet (§2).

Theorem 1 (chain-level metatheory). *In every reachable state:*

- (i) *Canonicity*: $\sigma = \{u \mapsto (\text{chain}(u), \text{char}_u)\}$ over survivors u ; the state is a function of the event set.
- (ii) *Birth constancy*: $\text{chain}(u)$ is fixed at u 's insert and never edited by any later transition.
- (iii) *Totality (no ties)*: *chain-lex* is a total order on every state's survivors.

Proof. (i) Survivorship is OR-set, hence a function of the event set, and a survivor's chain is determined by its insert event alone: by induction, **ins** writes $\text{chain}_\sigma(a) \cdot x = \text{chain}(a) \cdot x$ (the anchor is live at application, by honesty, and its stored chain is canonical), while **del** and the merge only delete or select records, never rewrite them. (ii) is (i) read off per node: no transition writes to an existing record. (iii) Distinct elements' chains are distinct (they end in distinct timestamps); if neither is a prefix of the other, the first difference compares two distinct timestamps in $\mathbb{N}$, and uniqueness of Lamport stamps is the no-tie property; prefix pairs are ancestor/descendant, ordered by the prefix rule. $\square$

The one-line separation, and the conservation law. The natural first attempt at tombstone-freedom is the *rehoming* RGA: keep flat (element, anchor) records, and let a delete splice its node out, re-parenting the orphans to the nearest live ancestor (the *climb*). We built and verified that design before this one (it is RA-linearizable, in the same Lean framework, §8), and it fails the L1 delete-reorder litmus (the $[b, a, c] \rightarrow [c, b]$ anomaly, Figure 1): the climb re-sorts a rehomed node by its own timestamp, so *its sort key is the birth chain truncated at each rehoming*. This design keeps the chain whole. That is the entire difference between the two datatypes, and it is the crisp form of the conservation law that our design sweep kept re-imposing: *the sort key must retain dead ancestors' timestamps*. Every design that forgets them has a named machine-checked countermodel (§7). What is negotiable is only how many bits the retained timestamps cost; that is the subject of §2, and the answer is: the log of each element's anchor gap, and nothing else.

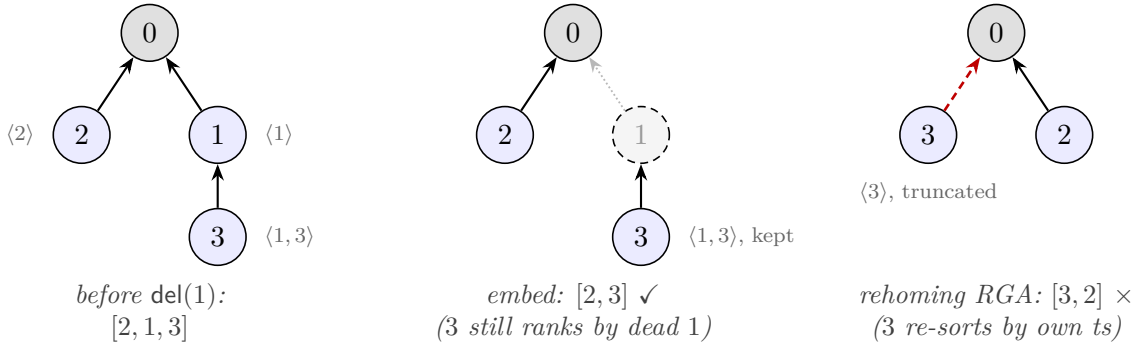


Figure 1: The one-line separation on the L1 delete-reorder litmus: insert 1 and 2 at the front, 3 after 1, delete 1. The rehoming RGA's climb (red) re-parents 3 and re-sorts it by its *own* timestamp: the truncated key $\langle 3 \rangle$ beats $\langle 2 \rangle$, and 3 jumps over 2. The embedded chain $\langle 1, 3 \rangle$ keeps ranking 3 by its dead parent's timestamp: $[2, 3]$, the delete-order-preserving read. Arrows point from a node to its anchor; dashed grey = dead.

Worked example (Figure 2). 6 and 10 concurrently insert at the front: chains $\langle 6 \rangle$ and $\langle 10 \rangle$; display $[10, 6]$, no arbitration, at every replica and merge order. Type 22, 16 under 6: chains $\langle 6, 22 \rangle$, $\langle 6, 16 \rangle$. Delete 6: the records of 22 and 16 are untouched and still carry 6's timestamp.

A third party concurrently inserts 8 at the front: $\langle 8 \rangle$ sorts below $\langle 10 \rangle$ and above $\langle 6, 22 \rangle$, so the display is $[10, 8, 22, 16]$ whether 8 arrives before or after 6’s death (machine-checked as the *credential countermodel*, the shape that kills every timestamp-oblivious variant, §4).

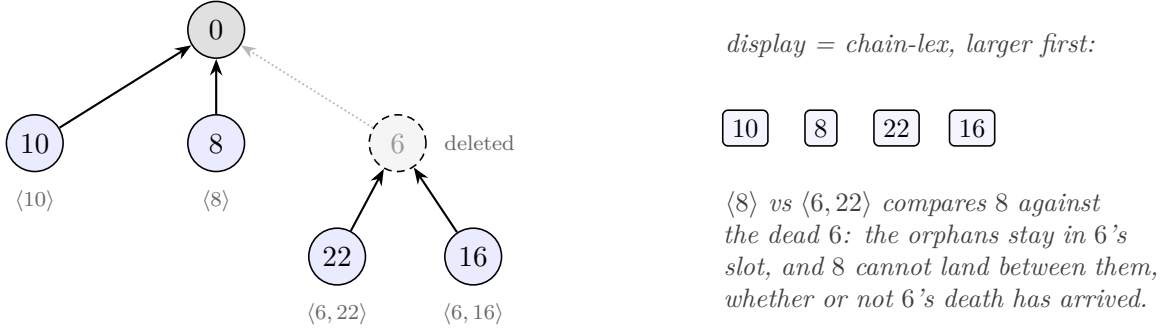


Figure 2: The credential countermodel as a birth tree. 6 and 10 race at the front; 22, 16 are typed under 6; 6 dies; 8 arrives at the front concurrently. Survivors sort by whole birth chains, so the read is $[10, 8, 22, 16]$ at every replica and merge order. Every timestamp-oblivious variant flips this shape (§7).

2 The representation: chains as entropy-coded coordinates

Everything above is the datatype; everything below is bits. A naive carrier would store a full timestamp at every chain level (each $\sim \log(\text{document age})$ bits, even for purely sequential typing) and compare whole chains at every read. This section stores the chains compactly without disturbing their order.

Vocabulary. A *codeword* is the bit string $C(\delta)$ encoding one chain level: one element’s timestamp, stored as its differential δ to its birth anchor (step 1 below). A *coordinate* is the bit string encoding a whole chain: the concatenation, root to element, of one block per level (the level’s codeword plus three framing bits fixed by the mint construction below), with nothing else marking the boundaries. We write $|s|$ for the length in bits of a bit string s . Read as a binary fraction, a coordinate names a dyadic interval in $(0, 1)$; codeword concatenation becomes interval nesting, and the interval is the $\alpha(u)$ of Theorem 2. Bit string and interval are two views of one object; we use whichever is clearer.

What the coding is for. Two jobs, only one of which is about correctness. *Job 1: cost.* The datatype layer already forced sort keys to retain dead ancestors’ timestamps (the conservation law, §1); the only remaining question is how many bits that retention costs. Without entropy coding, “tombstone-free” would be bookkeeping: a full timestamp per level grows with document age however calm the history. The *unary control* (**embed-tree**: the identical datatype with a unary code in the mint, codeword length linear in the value, kept as the experiment’s cost-control arm; step 2 and the entropy-law table) makes the gap measurable: 61 KB against the coded design’s 0.5 KB on the table’s 1000-element chain scenario. The coding is what makes the headline claim true: the dead survive as the entropy of the anchor gaps they were born across, and not more. *Job 2: faithfulness.* The chains must be stored so that plain lexicographic bit comparison reproduces chain-lex exactly, in every reachable state, forever. This is the single obligation the representation owes §1 (Theorem 2). Correctness lives upstairs; this layer is an implementation that must not lie.

Job 2 needs no entropy optimality at all: it needs exactly two structural properties of the per-level code, stated next.

The contract. The stored form must support four operations: *mint* (insert: produce the new codeword from the two timestamps alone, coordination-free, reading no siblings); *compare* (read: plain lexicographic comparison, with no tie rule); *fold* (delete: splice a dead node out by concatenating its block into each child’s stored string, exactly); and *refold* (merge: recompute survivors’ coordinates by the same concatenation). Each guarantee is bought by a specific property of C :

- **Monotone** ($\delta < \delta' \Rightarrow C(\delta) <_{\text{lex}} C(\delta')$) buys correct *compare*. A chain-lex first difference always compares two same-anchor entries, where timestamp order and δ order coincide (step 1); bitwise comparison of coordinates therefore equals timestamp comparison at the first differing level precisely when C is monotone.
- **Prefix-free** (no codeword is a prefix of another) buys three things at once. *Alignment*: in a comparison of two coordinates, the first differing bit falls inside the first differing level’s codewords, so levels cannot be confused across boundaries and no delimiters are needed; this is also what keeps *fold* and *refold* exact splices. *Geometry*: by the Kraft correspondence, prefix-free codewords occupy pairwise-disjoint dyadic cells, which is Theorem 2(i) and the reason a concurrent front-insert can never land inside a dead sibling’s span (Figure 4). *No ties*: cell disjointness plus Lamport uniqueness means sibling coordinates never tie, so the merge never decides an order and the read needs no tiebreak.
- **Universal, with a cheap floor** buys Job 1: C is defined for every $\delta \geq 1$, $|C(1)| = O(1)$ so continuation typing costs a constant per level, and $|C(\delta)| = \Theta(\log \delta)$ so a level pays the log of its anchor gap, whether the gap is a race or a cursor jump (step 1), and nothing else.

Monotone plus prefix-free is all of Job 2: any such code satisfies Theorem 2, which is why the Lean capstone is parametric in the code (§8) and why the unary control proves the same theorems at absurd cost. Entropy coding enters only in *which* monotone prefix-free code is chosen; the choice sets the constant in the bits-per-level bill and nothing in the correctness story. One requirement cuts across all four operations: the arithmetic is exact, never rounded (see the binary64 refutation below).

Step 1: differential coding. By causality $\delta = x - a \geq 1$. A chain-lex first difference always compares two entries with the *same* anchor (the shared prefix), and for a fixed anchor, ordering by x and by δ coincide, so each chain level may store the differential instead of the timestamp. Call δ the *anchor gap*: the events elapsed between the anchor’s mint and the insert’s. Continuation typing (each character anchored at its predecessor) has $\delta = 1$ at every level regardless of how long the document has lived. Two things widen the gap, and they pay the same bill: a genuine race with timestamp gap g costs $\log_2 g$ bits, and so does a *cursor jump*, the first character typed at an old anchor; its successors anchor at fresh characters and are back to $\delta = 1$. The differential is an encoding choice, not semantics: the mathematical object remains the timestamp.

Step 2: entropy coding. For $\delta \geq 1$ let L be the bit-length of δ and

$$C(\delta) = \underbrace{1 \cdots 1}_{L-1} 0 \text{ } (\delta \text{ with its leading bit removed}) \quad (|C(\delta)| = 2L - 1),$$

an order-preserving prefix-free code: exactly the contract. The length accounting: prefix-freedom makes each codeword self-delimiting, and a self-delimiting code over unbounded δ must spend bits twice, once on the value and once announcing its own length. C pays each bill at $\sim \log_2 \delta$. The header $1^{L-1}0$ is a unary announcement “ $L-1$ payload bits follow” (L bits); the payload is δ ’s trailing $L-1$ bits, since the leading bit of an L -bit number is always 1: once the header has said L it carries no information and is dropped. Hence

$$|C(\delta)| = \underbrace{(L-1)+1}_{\text{header}} + \underbrace{(L-1)}_{\text{payload}} = 2L - 1 = 2\lfloor \log_2 \delta \rfloor + 1,$$

e.g. $C(1) = 0$, $C(2) = 100$, $C(3) = 101$, $C(5) = 11001$. This is Elias γ with its header flipped: γ announces the length in *zeros*, which is prefix-free but order-reversing across length classes: at the first divergence the longer, hence larger, number shows a header 0 against the shorter’s leading payload bit 1. Writing the header in ones makes that same position decide in favor of the longer code, so lexicographic order on codewords is numeric order on δ . The γ -flip is the building block, not the last word: only the *value* bill is forced at $\log_2 \delta$, and the *length* bill can itself be entropy-coded. Applying C to the length field (then the payload) gives the order-preserving flip of **Elias** δ , prefix-free and monotone by the same across-length-class argument, at $\log_2 \delta + O(\log \log \delta)$ per level; *this is the canonical mint*. Any order-preserving prefix-free code works here, and the artifact exists in three proved encodings of one datatype (**embed-tree** unary, **embed-code** the γ -flip, **eliasDeltaCode** the δ -flip); the capstone theorem is inherited at each verbatim, with zero new proof content (§8): the code-parametricity claim, machine-validated. Two honest footnotes: per level the δ -flip wins only from $\delta \geq 32$ and loses at $\delta \in \{2, 3\} \cup [8, 15]$ (kernel-checked length comparisons), and measured editing tails are thin, so on real traces the two flips are a wash; the dominance shows in race-heavy shapes (the entropy-law table below) and in the asymptotics.

The mint. Writing $b = 1 \cdot C(\delta)$ (a leading 1 keeps coordinates positive) and $w = 2^{-|b|-2}$, the mint is the second quarter of b ’s dyadic cell:

$$M(\delta) = (0.b + w, 0.b + 2w) \subset [0.b, 0.b + 4w).$$

The quarter placement leaves a gap below and headroom above inside the cell, and the open space $(0.b + 2w, 0.b + 4w)$ is never minted: it exists so that the interval has room to *contain* the node’s own subtree without touching the next cell. This also pins down the per-level *block* promised in the vocabulary: $0.b + w$ written as a string is $b01$, so the block is $1C(\delta)01$, the codeword in its three framing bits (the leading 1 for positivity, the 01 that selects the second quarter). The block is what a node stores; read as a binary fraction it is the lower endpoint of $M(\delta)$, and flipping the trailing 01 to 10 gives the upper. Worked out for small deltas:

δ	$C(\delta)$	block = $1C(\delta)01$	$M(\delta)$	bits/level
1	0	1001	(9/16, 10/16)	4
2	100	110001	(49/64, 50/64)	6
3	101	110101	(53/64, 54/64)	6
4	11000	11100001	(225/256, 226/256)	8
5	11001	11100101	(229/256, 230/256)	8
10	1110010	1111001001	(969/1024, 970/1024)	10
100	1111110100100	1111111010010001	(65169/65536, 65170/65536)	16

Three things to read off. The endpoints are consecutive dyadics at scale $2^{-(|b|+2)}$ (numerators n and $n+1$): the mint is one tick wide at its own scale, with the tick below it (the gap) and the two

ticks above it (the subtree headroom) never minted. The rows are strictly increasing as fractions: monotonicity made visible. And the bit count is $2L + 2$ for an L -bit δ : four bits at $\delta = 1$ (the sequential-typing constant of the entropy law), sixteen at $\delta = 100$.

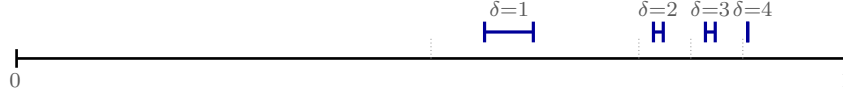


Figure 3: Mints $M(\delta)$ for $\delta = 1..4$ inside the parent's unit interval (cell boundaries dotted). Larger deltas sit strictly higher; distinct deltas are disjoint with gaps; each mint is a quarter of its cell, leaving room for the node's own subtree. Drawn for the γ -flip C' ; the construction reads verbatim over any order-preserving prefix-free code, in particular the canonical Elias- δ mint (step 2).

The stored state. The carrier stores each live node parent-relatively:

$$\sigma : t \mapsto (\text{par} \in \mathbb{N}, \quad s \in \{0, 1\}^*, \quad \text{char}),$$

where s is a block ($1C(\delta)01$ at birth; after folds, below, a concatenation of blocks) *relative to the parent*, and $\text{par} = 0$ denotes the root sentinel (which owns $(0, 1)$ and is never stored). The record holds no interval; the interval is the string's decoding,

$$(lo, hi) = (0.s, \quad 0.s + 2^{-|s|}) \subset (0, 1),$$

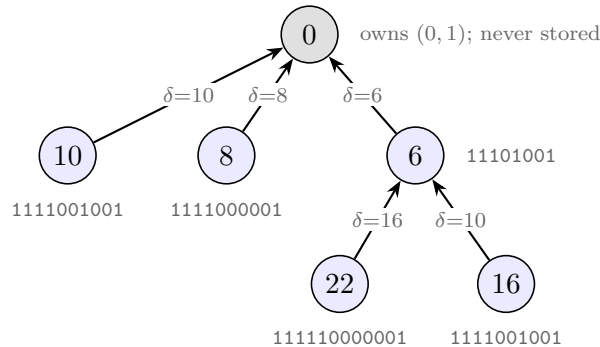
consecutive dyadics at s 's own scale, one block at a time exactly as in the mint table. The maintenance below and Theorem 2 are stated on the intervals, and every operation they perform on an interval is a string operation on s : composition is concatenation, comparison is lexicographic. (The validation model stores the decoded pair directly, as exact fractions.) A node's *absolute* interval is the affine composition of the intervals along its parent chain. For a node v with mint $M(\delta_v) = (lo_v, lo_v + w_v)$, let

$$\varphi_v : (0, 1) \rightarrow M(\delta_v), \quad \varphi_v(x) = lo_v + w_v \cdot x,$$

the unique increasing affine bijection of the unit interval onto v 's mint; it maps a subinterval (a, b) to $(lo_v + w_v a, lo_v + w_v b)$, and on strings it prefixes v 's block: $\varphi_v(0.y) = 0.(s_v y)$. Write

$$\alpha(u) = \varphi_{c_1} \circ \dots \circ \varphi_{c_k}(M(\delta_u))$$

for the composition along u 's birth chain $c_1 \dots c_k u$ (δ_v is v 's delta to its birth parent): $\alpha(u)$ is the *encoding of chain(u)*, and at the string level it decodes the concatenation of the blocks along the chain. Worked on the tree of Figure 2, redrawn before $\text{del}(6)$ with the delta (child minus anchor) on each edge and the stored block under each node:



Writing $(n, n+1)/2^k$ for the interval $(n/2^k, (n+1)/2^k)$ and $\approx$ for its lower endpoint:

node	δ	stores (par, s)	decode of s	α (absolute)	$\approx$
10	10	(0, 1111001001)	$(969, 970)/2^{10}$	$(969, 970)/2^{10}$	0.94629
8	8	(0, 1111000001)	$(961, 962)/2^{10}$	$(961, 962)/2^{10}$	0.93848
6	6	(0, 11101001)	$(233, 234)/2^8$	$(233, 234)/2^8$	0.91016
22	16	(6, 111110000001)	$(3969, 3970)/2^{12}$	$(958337, 958338)/2^{20}$	0.91394
16	10	(6, 1111001001)	$(969, 970)/2^{10}$	$(239561, 239562)/2^{18}$	0.91385

The root's children store their absolute intervals already ($\text{par} = 0$). The grandchildren compose: 22's absolute string is its parent's block followed by its own, 11101001 111110000001 (20 bits, hence the 2^{20}), and likewise 16's (18 bits). Both absolute intervals sit inside $M(6) = (0.91016, 0.91406)$ while 8's sits above it, so the descending- α read is $[10, 8, 22, 16]$ and 8 cannot land between the orphans. Nodes 16 and 10 store the same string, both being $\delta = 10$ births; records are parent-relative, so this is no collision. After $\text{del}(6)$ the fold rewrites the two records to ($\text{par } 0$, the concatenated s) and the α column is unchanged bit for bit. Figure 4 draws these intervals (widths schematic).

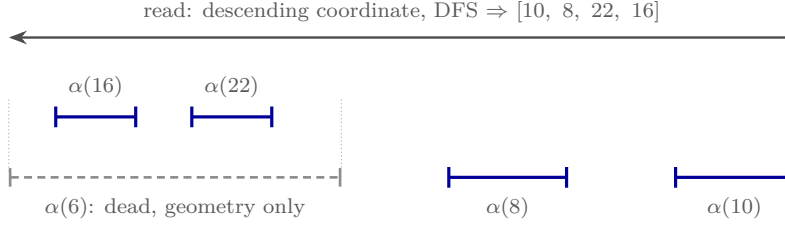


Figure 4: Figure 2 downstairs (widths schematic; true mints are the dyadic cells of Figure 3). Deleting 6 removes its record, but $\alpha(16)$ and $\alpha(22)$ are compositions *through* $M(6)$: the dead node survives as the dashed container, the tombstone compressed into geometry. 8's interval can never land inside $\alpha(6)$'s span: distinct deltas occupy disjoint cells (Theorem 2(i)), which is the credential verdict, now a fact about intervals.

Maintenance. How the stored form tracks the datatype:

- $\text{ins}(x \leftarrow a)$ mints $M(x - a)$ relative to a : the new record depends only on the two timestamps.
- $\text{del}(d)$ *isometrically folds* d into its children: each child c is re-parented to d 's parent and d 's record is substituted into c 's. On the stored strings this is concatenation, $c.s := d.sc.s$; read as intervals, $c.\text{frac} := d.\text{frac} \circ c.\text{frac}$, where $I \circ J$ is the image of J under the increasing affine bijection of $(0, 1)$ onto I (the φ of the stored state, with I in place of a mint): for $I = (lo, lo + w)$,

$$I \circ J = (lo + w \cdot lo_J, lo + w \cdot hi_J).$$

The substitution is exact. Upstairs, deletion changes nobody's sort key (§1); the fold is the parent-relative storage scheme *re-normalizing* while every absolute interval stays fixed (Figure 5): representation maintenance, not datatype semantics. On the running example, $\text{del}(6)$ removes 6's record and substitutes it into both children: 22's record (6, 111110000001) becomes (0, 11101001 111110000001), whose decode is

$$(233, 234)/2^8 \circ (3969, 3970)/2^{12} = (233 \cdot 2^{12} + 3969, 233 \cdot 2^{12} + 3970)/2^{20} = (958337, 958338)/2^{20},$$

and likewise 16's becomes $(0, 11101001\ 1111001001)$, decoding $(239561, 239562)/2^{18}$. Both are exactly the α column of the worked table: the absolute intervals did not move, only the parent-relative bookkeeping did.

- The merge *refolds*: for each survivor, compute its absolute interval by folding its record to the root inside the input that holds it (a record's stored parent is always live in its own input, so the chain stays in one input; by Theorem 2(iii) the choice of input cannot matter, asserted at runtime and never fired); re-parent to the nearest *surviving* ancestor; re-relativize.
- The read is depth-first from the root, children in descending order of *lo*. No tie rule is needed (Corollary 1).

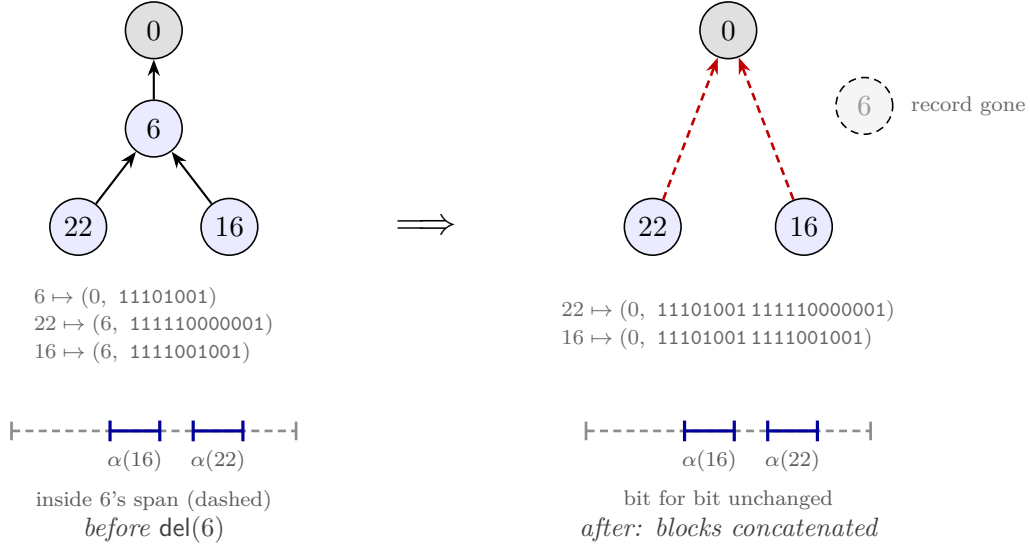


Figure 5: The isometric fold on the running example. $\text{del}(6)$ deletes 6's record and concatenates its block 11101001 onto the front of each child's, re-parenting both to the root (red: the re-parenting, a storage move only). The stored states change: 22 and 16 now carry two-block strings and par 0. The absolute intervals $\alpha(22) = (958337, 958338)/2^{20}$ and $\alpha(16) = (239561, 239562)/2^{18}$, and with them the display, are bit-for-bit unchanged: upstairs, the sort keys still read $\langle 6, 22 \rangle$ and $\langle 6, 16 \rangle$. The dashed span is 6's old interval, surviving as pure geometry.

Why the codeword must announce its own end. Before the theorem, the countermodel that motivates it. A single codeword in a field of known extent needs no header, and the length looks recoverable from the string itself; plain binary shows what breaks, once per purchase in the contract. *Geometry*: $B(2) = 10$ prefixes $B(5) = 101$, so $B(5)$'s cell sits inside $B(2)$'s; containment means ancestry, so the $\delta=5$ sibling lands amid the $\delta=2$ sibling's descendants and non-interleaving dies with everything alive. *Parsing*: once a fold erases the structural boundary between blocks, $101100\dots$ reads as $B(2)$ then $11\dots$ or as $B(5)$ then $00\dots$, and nothing outside the string can say. *Ties*: a binary fraction cannot see trailing zeros, so 10 and 100 both denote $\frac{1}{2}$; distinct siblings mint touching intervals, and the id tiebreak wakes into an arbiter that orders by id, not chain-lex. Nor can the header bits be dodged: prefix-freedom demands $\sum_{\delta} 2^{-\ell(\delta)} \leq 1$, and plain binary puts 2^{L-1} values at every length L , so every prefix-free code pays a length announcement of order

$\log L$ somewhere (out-of-band boundary tables cost the same bits and forfeit the string as a self-contained key; database key encodings inline it for the same reason). And the fold names where the boundary went: the tree node that carried it structurally is the deleted tombstone, whose ordering verdict becomes the payload and whose boundary becomes the header. “Tombstone compressed into coordinate entropy” is literal at the bit level.

The one obligation. The four theorems of the earlier drafts collapse into a single statement about the encoding.

Theorem 2 (encoding faithfulness). *(i) Order-embedding at one level: for $\delta \neq \delta'$, $M(\delta)$ and $M(\delta')$ are disjoint with a gap; $\delta < \delta' \Rightarrow M(\delta)$ lies entirely below $M(\delta')$.*

(ii) Homomorphism: α realizes chain concatenation as affine composition and embeds chain-lex in the interval order: for distinct u, v , if $\text{chain}(u)$ is a proper prefix of $\text{chain}(v)$ then $\alpha(v) \subset \alpha(u)$ (and, if both are live, the two are never read-time siblings); otherwise $\alpha(u) \cap \alpha(v) = \emptyset$, with the chain-lex-greater element’s interval strictly higher.

(iii) Exact maintenance: in every reachable state, every live node’s stored records compose to exactly $\alpha(u)$.

Proof. (i) C is prefix-free: codes of lengths $2L-1 < 2M-1$ differ at position L (the shorter’s header terminator 0 against the longer’s header 1); equal-length codes differ in the payload. Prefix-free codes have pairwise-disjoint dyadic cells. C is monotone as a binary fraction: within a length class the payload is δ ’s low bits in order; across classes the position- L bit decides in favor of the longer code. Disjoint cells ordered by their codes, and mints interior quarters of their cells, give both claims. (ii) If neither chain is a prefix of the other, they first differ on distinct deltas relative to the *same* birth ancestor (§2, step 1); (i) separates them there, and affine images of disjoint intervals are disjoint and order-preserved (each φ is increasing). If $\text{chain}(u)$ prefixes $\text{chain}(v)$, then $\alpha(v) \subseteq \varphi_{c_1} \circ \dots \circ \varphi_{c_k}(\varphi_u((0, 1))) = \alpha(u)$ by construction; and if both are live, v ’s nearest live ancestor is u or deeper, so they never share a current parent, and sibling groups are pairwise disjoint. (iii) By induction over transitions. *ins* establishes it by definition. *del*’s fold replaces (par, frac) pairs by their exact composition, leaving α unchanged. The merge recomputes precisely $\alpha(u)$ (folding a record chain to the root) and re-relativizes against another α ; ties cannot exist by (i) and (ii), so no step of any transition ever rewrites an interval by anything but exact composition. $\square$

Corollary 1 (display $\equiv$ chain-lex; no tie rule). *The read (DFS by descending lo) enumerates survivors in exactly chain-lex order: per level, larger delta first, so among same-parent siblings, exactly newest first; survivors rank by their own timestamps, folded heirs by their dead founder’s. Sibling lo-values never tie; the read’s id tiebreak is dead code. Machine checks: pairwise sibling disjointness asserted at every read of 320 randomized executions, all clean; over 19,531 multi-member sibling groups, own-timestamp order violated 3,481 times (exactly the post-fold heir groups; forcing own-ts there is the refuted dissolution behavior), **chain-lex violated 0 times**.*

The entropy law. A coordinate is the concatenation of the codes along the birth chain: $\Theta(\sum_i \log \delta_i)$ bits, the entropy of the chain. Continuation typing ($\delta = 1$ at every level) costs ~ 4 bits per level under every code here; a level minted across an event gap g , by a race or by the first character after a cursor jump, costs $\log_2 g + O(\log \log g)$ bits under the canonical Elias- δ mint ($2\log_2 g + 1$ under the γ -flip). *The state pays for the magnitude of edit non-locality, concurrent or not, and for nothing else.* Measured (1000 elements; the numbers are order metadata alone,

survivors’ coordinate bits read off the dyadic denominators; characters are excluded, since data costs are identical across all designs):

scenario	quarter carve (refuted)	unary mint	γ -flip mint	Elias- δ mint
chain then delete-all ($\delta=1$)	2^{2001}	2^{502501} (~ 61 KB)	2^{4001} (~ 0.5 KB)	2^{4001} (~ 0.5 KB)
1000 root siblings ($\delta=t$)	— (ties)	$\max 2^{1003}, 125$ KB	$\max 2^{23}, 5$ KB	$\max 2^{20}, 4$ KB

The carrier. The natural representation is a **parent-relative bit string per node**: the fold is concatenation, comparison is lexicographic ($O(1)$ -ish with structure sharing); the rationals are the semantics of the strings, kept for the proofs. Nothing here computes $M(\delta)$: the endpoints $0.b + w$ and $0.b + 2w$ are the strings $b\ 01$ and $b\ 10$, so the mint emits a block and the fold concatenates, and *hi* need not be stored at all. The Lean mechanization drops even the framing bits: its coordinate is the bare codeword concatenation on `List Bool` (`mint` appends $\Gamma.\text{enc}(t - a)$ to the anchor’s coordinate), with the prefix rule stated directly on strings; the quarter framing serves the interval realization alone, buying positivity and subtree interiority. The rational $M(\delta)$ is computed in exactly one place, the validation model `embed_tree.py`, as exact fractions; the entropy tables read their bit counts off those denominators. IEEE 754 binary64 is machine-refuted as the carrier, a faithfulness failure rather than a design failure: mints degenerate at $t = 52$, depths underflow at 44, and rounding voids Theorem 2(iii) (6/120 PBT executions die). The obstruction is a pigeonhole (chains carry $\Theta(\sum \log \delta)$ bits; a fixed-width word holds 64), but binary64 is sound as a *comparison cache* with exact fallback on float equality.

Timestamps in practice. The unique stamps in $\mathbb{N}^+$ are Lamport’s 1978 construction, flattened: each replica ticks a counter per local operation, advances it to the max on merge, and breaks ties by replica id, packed as $t \cdot 2^k + r$ with k id bits. Uniqueness and causality ($t_a < t_x$ gives $\text{packed}_a < \text{packed}_x$ regardless of ids, so $\delta \geq 1$) both survive the packing; the proof layer only *assumes* uniqueness (the Lamport-structural part of honesty), and this construction is the witness that the assumption is realizable. Two entropy caveats. (i) Packing inflates every delta by 2^k , costing $\sim 2k$ extra bits per chain level, which turns the 0.5 KB sequential run above into ~ 3 KB at $k = 10$. Cheaper: extend the mint’s code to the pair, $C(\Delta t)$ followed by a k -bit id. The pair code is still order-preserving prefix-free (lex on $(\Delta t, r)$), and costs k bits only where the level is minted. A further option, unvalidated against the battery and recorded as a design note, is to order sibling ids *relative to the anchor’s* (any deterministic per-anchor id order is a legitimate tiebreak; convergence needs agreement, not a uniform order), making “same author as my anchor”, i.e. sequential typing, cost ~ 1 bit, so only genuine cross-author races pay $\log R$. (ii) The clock must be *dense logical time*, *never physical*: wall-clock or HLC stamps make δ measure elapsed time rather than elapsed events: a keystroke every 100 ms over μs -resolution stamps mints $\delta \approx 10^5$, ~ 34 bits per level for a purely sequential edit. The entropy law holds precisely because the counter ticks once per event, making δ an event-count gap; if wall-clock order is ever wanted, it belongs in a display-layer tiebreak, never in the minted coordinate.

What remains open. Dead chain entries concatenated into survivors’ strings are the conservation law’s irreducible content (§1); reclaiming them under *causal stability* (renumber once every replica has seen the delete, the only condition under which rewriting geometry is safe, since no verdict involving the dead can be contested again) is the deepest of the open cost items, and it is

orthogonal to correctness. The wider compression headroom is catalogued with its research questions in the companion assessment `whiteboard/order-coding-compression-notes.md`: context-conditioned per-anchor codes (licensed by comparison locality: a chain-lex first difference never crosses anchors), coalescing $\delta=1$ runs, Steiner-sharing dead prefixes across co-heirs, stability-gated rank re-coding, and the Elias- δ constant (step 2 above).

3 The insert prefix: for the proof, and the proof alone

The insert payload π exists for verification, not execution. This section says why it must exist, and why an implementation may drop it from the wire.

Why rc must be Either. The verification framework (§8) resolves a concurrent pair of non-commuting operations by consulting a declared conflict-resolution order rc ; any *oriented* entry for an insert/delete pair, however, incurs a base commutation obligation that is machine-refuted for this design space: ordering $ins(x \leftarrow a)$ against $del(a)$ (in either direction) is undischargable as a conflict entry, because rehoming is non-transitive across chains of deletions. So $rc = \text{Either}$: the proof must show the two *commute*, in both orders, on all applicable states.

The commutation, and where the prefix earns its keep. Order 1, $ins(x \leftarrow a)$ then $del(a)$: x is born with chain $\pi \cdot x$, and the delete touches no chain (downstairs: the fold re-homes x 's record with composed coordinates; by exactness $\alpha(x) = \alpha(a) \circ M(x - a)$). Order 2, $del(a)$ then $ins(x \leftarrow a)$: a is gone; application must still be *total* and land x on the same sort key. This is what the carried prefix $\pi = \text{chain}(a)$ (equivalently, the anchor's coordinate) provides: apply sets $\text{chain}(x) = \pi \cdot x$ (downstairs: $\alpha(x) = \pi \circ M(x - a)$, attached to the nearest live ancestor). The two orders produce *equal* states: commutation to equality, not merely up to $\approx$, courtesy of canonicity (Theorem 1). Figure 6 draws the square on the smallest instance.

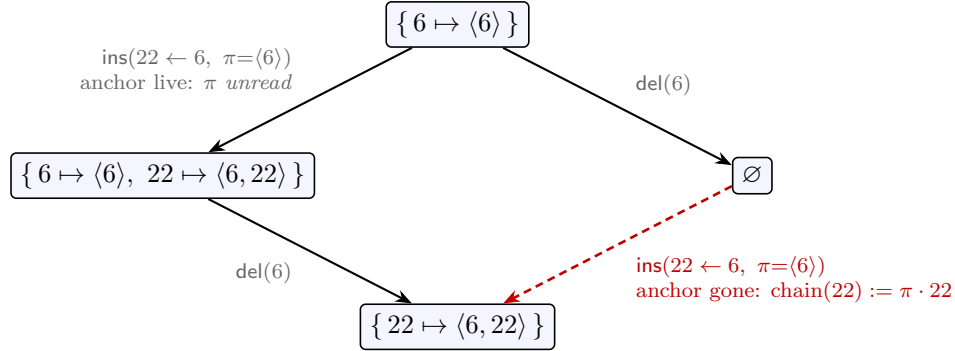


Figure 6: $rc = \text{Either}$ discharged: $ins(22 \leftarrow 6)$ and $del(6)$ commute *to equality*. Down the left, the anchor is live and application never consults π (the reachable, honest order). Down the right, 6 is already gone: the state cannot supply the anchor's chain, so the carried $\pi = \text{chain}(6)$ does, landing 22 on the same sort key. The right-hand state is unreachable under honesty; the verification condition quantifies over it anyway, and π exists exactly to make that corner total.

Ghost, in the strict sense. On every reachable state the anchor is live at application (honesty), and the application reads only the state and the two timestamps: π is never consulted. The Lean artifact carries this as `ins_prefix_ghost` (`Sal/MRDTs/RGA_Embed/RGA_Embed_MRDT.lean`):

on accurate ops the always- π transition and the state-reading transition are literally the same function: the two conventions of this section coincide on every honest state. An implementation may therefore drop π from the wire entirely; the *proof* may not, because the verification conditions quantify over reorderings that visit unreachable states, and there application without π is partial. The prefix costs $O(\sum \log \delta)$ bits, the same entropy as the coordinate itself, and exists at the proof layer alone.

4 Validation

All numbers reproduce from `whiteboard/litmus/` in the accompanying Sal repository (`embed_tree.py`; harnesses `litmus.py`, `pbt.py`). Notation: $\text{RGA}_\dagger$ is the published tombstoned RGA (Roh et al.; §7); the *rehoming* RGA is the convergence-verified flat tombstone-free design of §1 (since demoted repo-wide: the L1 row below was generalized into a kernel-checked refutation of its sequential specification, `rehoming_seq_refuted`, and this design took the canonical seat; §8). The L1–L25 battery is the litmus suite accumulated over this project’s design sweep; each entry is the named countermodel that killed at least one of 16 prior candidate designs. The one exception below, L19, is discharged by the sided generalization of §5.

check	verdict
litmus battery L1–L25 (incl. every countermodel that killed 16 prior designs: ceiling escapes, tie inheritance, three- branch topology, frame mixing, fold-verdict inheritance)	clean, except one-sided L19 (backward-run interleaving, the published RGA’s own anomaly)
credential countermodel (tie heirs vs mid-ts third party, death-before vs death-after topologies)	converges, no flips; = $\text{RGA}_\dagger$ ’s reads
subordination-escape and retroactive-subordination CEs	survive
randomized version-DAG PBT (FLIP / CONV / LIVE / DUP)	120/120 and 300/300 clean
per-read sibling-disjointness assertion sweep	320/320 clean
chain-lex instrumentation (19,531 sibling groups)	0 violations
lockstep vs the published tombstoned RGA	read-equal at every apply/merge/read, 120/120
L1 delete-reorder vs the verified rehoming RGA	embed [2, 3] ✓ vs rehoming [3, 2] ×

The lockstep row is the identification: *the embedded-chain RGA is observationally the published tombstoned RGA with the tombstone set compressed into coordinate entropy*, machine-checked on 120 executions and now a kernel-checked theorem (the compaction theorem, §8): on every honest event set, any enumeration of the tombstoned RGA’s visible ids in its own visible order *equals* the embed read, element for element. The L1 row is the separation from the verified rehoming RGA, and it is exactly the one-line difference of §1: that design’s climb re-sorts a rehomed node by its own timestamp (the $[b, a, c] \rightarrow [c, b]$ anomaly), i.e. its sort key is the birth chain *truncated at each rehoming*; this design keeps the chain whole.

5 The sided generalization: two-sidedness as a parameter

The one blemish in the battery is L19: two replicas that each type a *backward* run at the same place (repeatedly “insert before the current head”) interleave, because every insert-before anchors at the same left neighbor and both runs land in one sibling group. The fan is not a traversal artifact: no read order of the same birth tree un-fans a fan, so any fix must change where insert-before

anchors. The Fugue family’s answer is a second side: insert-before anchors at the *right* neighbor as an L-entry, so a backward run is a chain, exactly as a forward run already is.

The L19 example, concretely. It is the battery’s own trace (Figure 7). The LCA document is $\langle 1 \rangle$ (one insert, stamp 1, anchored at the start marker $\diamond$). Then, concurrently, replica A types a backward run at the head, three inserts each anchored “before the current first element”:

$$\text{ins}(10 \text{ at head}), \quad \text{ins}(30 \text{ at head}), \quad \text{ins}(50 \text{ at head}),$$

so A locally reads $\langle 50, 30, 10, 1 \rangle$; replica B does the same with stamps 21, 41, 61 and locally reads $\langle 61, 41, 21, 1 \rangle$. (The stamps interleave numerically, $10 < 21 < 30 < 41 < 50 < 61$, because the two replicas’ Lamport clocks advance together.) In the one-sided datatype, “insert at the head” can only anchor at the left neighbor, which is $\diamond$ for every one of the six inserts: all six become R-children of $\diamond$, one sibling group, and the sibling tiebreak (newest first) merges the two runs by stamp:

$$\langle 61, 50, 41, 30, 21, 10, 1 \rangle$$

after the merge, the two sentences shuffled together. No traversal of that tree can do better; the information “these three form one run” is simply not in a sibling fan whose stamps interleave.

The sided datatype, and why it prevents the interleaving. The generalization does not build a second datatype. An insert names an anchor *and a side*; chains become sequences of (side, timestamp delta) entries; and the display rule generalizes from the prefix rule (“a prefix precedes its extensions”) to the in-order rule (“L-extensions, then the node, then R-extensions”). One marker trick makes this ordinary lexicographic comparison again: a node’s sort key is its coordinate followed by a virtual terminator, and the sided symbol alphabet is ordered

$$(R, 1) < (R, 2) < \dots < \text{marker} < \dots < (L, 2) < (L, 1),$$

with the L band *mirrored* (among L-siblings the newer sits adjacent to the node). Marker above the R band puts a node before its R-extensions; marker below the L band puts L-extensions before the node.

On the example, the Fugue anchoring rule (the generation-policy paragraph below) sends each “insert at head” to the *successor* as an L-entry, so each run is an L-chain (Figure 7, right). Writing chains as entry sequences with deltas relative to the parent’s stamp ($10 = 1 + 9$, $30 = 10 + 20$, ...):

$$\begin{array}{ll} \text{chain}(1) = (R, 1) & \text{chain}(10) = (R, 1) (L, 9) \\ \text{chain}(30) = (R, 1) (L, 9) (L, 20) & \text{chain}(21) = (R, 1) (L, 20) \\ \text{chain}(50) = (R, 1) (L, 9) (L, 20) (L, 20) & \text{chain}(41) = (R, 1) (L, 20) (L, 20) \end{array}$$

and the marker-lex comparisons (keys descending in display order) work out as follows, writing M for the marker:

- *L-extensions precede the node:* $\text{key}(10) = (R, 1)(L, 9)M$ vs $\text{key}(1) = (R, 1)M$ first differ at position 2, $(L, 9)$ vs M , and the L band sits above the marker: so 10 displays before 1. Likewise 30 before 10, 50 before 30: each backward keystroke lands immediately left of the previous one, which is the typed intent.
- *The two runs are decided once, not per pair:* $\text{key}(10)$ vs $\text{key}(21)$ first differ at position 2, $(L, 9)$ vs $(L, 20)$, and the L band is mirrored, so $(L, 20) < (L, 9)$: A’s run displays before

B's. Now *every* node of A's chain carries the prefix $(R, 1)(L, 9)$ and every node of B's carries $(R, 1)(L, 20)$, so *every* cross-pair first-differs at that same position 2 with that same verdict. The blocks cannot interleave:

$$\langle 50, 30, 10, 61, 41, 21, 1 \rangle.$$

This is the datatype-level theorem **schain_subtree_convex** (a subtree is an in-order interval; §8): *any* policy that keeps a run inside one subtree gets non-interleaving from the datatype, for free.

The current datatype is the all-R fragment of this order, *exactly*: with L uninhabited the in-order rule degenerates to the prefix rule, and the existing key of §2 (the sym-mapped bits plus terminator) is literally this construction with the L band empty. That is a theorem, not a slogan: the mechanization's fragment theorem (§8) proves all-R chains order exactly as one-sided chains.

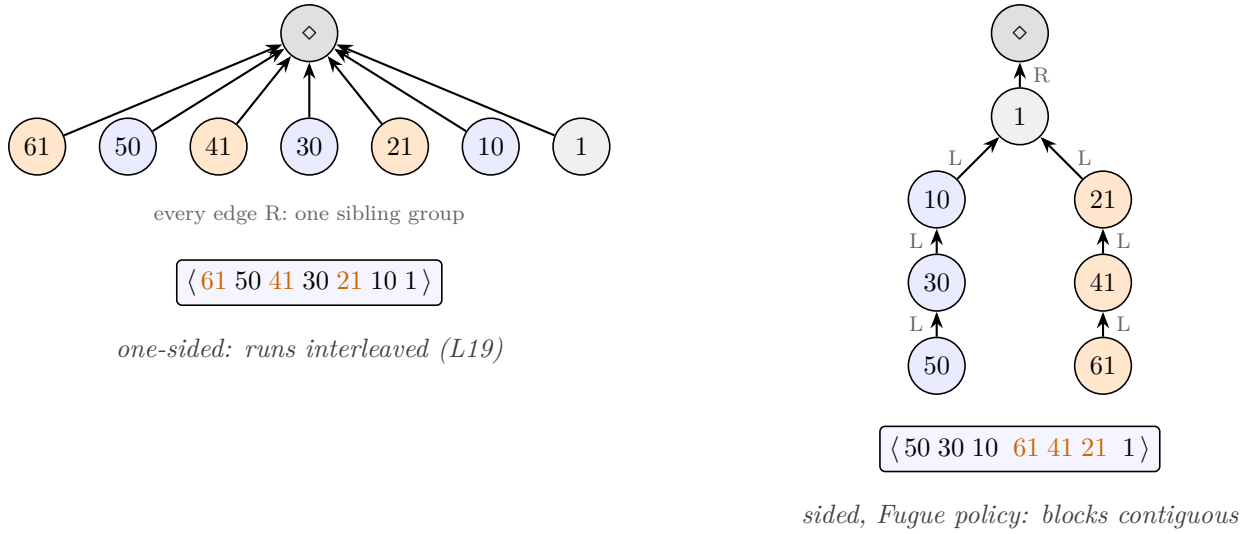


Figure 7: L19 under the two anchoring rules, the battery's own trace. LCA document $\langle 1 \rangle$; replica A (blue) types backward 10, 30, 50 before the head while replica B (orange) concurrently types backward 21, 41, 61. One-sided (left): every insert-before anchors at the same left neighbor $\diamond$, both runs land in one R-sibling group, and the timestamp-sorted fan interleaves them. Sided under the Fugue policy (right): insert-before anchors at the *successor* as an L-entry, each run is an L-chain hanging off 1, and the in-order read keeps the blocks contiguous: L-extensions precede the node (50, 30, 10 before their parents' line ending at 1), and among the L-siblings 10, 21 of node 1 the older run displays first (the L band is mirrored). Both display strips are machine outputs of `embed_sided.py`.

How the encoding realizes it, for zero extra bits. The one-sided block of §2 spends a constant leading 1 per level whose only job is keeping coordinates nonempty; repurpose it as the side bit. The L band needs an order-reversing prefix-free code, and the bit complement $\overline{C}$ is exactly that at identical lengths (flipping every bit flips every first-difference verdict and preserves both lengths and prefix relations), so the block becomes

$$\text{block}(\text{side}, \delta) = \text{sideBit} \parallel (C(\delta) \text{ if } R, \overline{C}(\delta) \text{ if } L),$$

prefix-free per band, with no framing change and no new bits. On the example, with the binary delta code ($C(1) = 0$, $C(9) = 1110001$, $C(20) = 111100100$, so $\overline{C(9)} = 0001110$, $\overline{C(20)} = 000011011$), the stored coordinates are the concatenated blocks:

node	chain	stored coordinate (side bits marked)
1	$(R, 1)$	<u>1</u> 0
10	$(R, 1)(L, 9)$	<u>1</u> 0 00001110
21	$(R, 1)(L, 20)$	<u>1</u> 0 0000011011
30	$(R, 1)(L, 9)(L, 20)$	<u>1</u> 0 00001110 0000011011
50	$(R, 1)(L, 9)(L, 20)(L, 20)$	<u>1</u> 0 00001110 0000011011 0000011011

To compare two coordinates, parse: read a side bit; if 1, decode one C codeword as an R-band symbol (R, δ) ; if 0, decode one $\overline{C}$ codeword (decode C on the flipped bits) as an L-band symbol (L, δ) ; repeat, then terminate with the marker. Prefix-freeness per band delimits every block, so no length fields or framing are needed, exactly as one-sided. The complement is what makes the mirror free: within the L band, plain descending comparison of the *stored* bits is already the mirrored delta order ($\overline{C(20)} = 000011011 < \overline{C(9)} = 0001110$ bitwise, matching $(L, 20) < (L, 9)$), so the carrier stores entropy payload only and the marker-lex of the previous paragraph is read off the bits. Faithfulness of this parse (unique decodability plus order embedding of the sided alphabet) is machine-checked by the chain-model/code-model lockstep below.

Two honest cost notes. Runs absorb the side bit like every other run-constant (paid once at the head). On *this* trace the sided coordinates are longer than the one-sided fan's (the runs' deltas are 20 because the two replicas' Lamport clocks interleave); the compression claim is about the common case, continuation-stamped backward typing ($\delta = 1$ per keystroke, an $0\|\overline{C(1)} = 01$ block per level), where today's design pays a growing- δ fan. Two-sidedness buys the semantics always and the compression exactly when backward runs are locally stamped.

Side selection is a generation policy, not a datatype choice. If insert merely “takes a side,” convergence and RA-linearizability hold for any sides whatsoever: those proofs never consult the order's intent. What depends on *which* side is picked is the ordering intent. Fugue's non-interleaving needs the rule “right child of the left neighbor when it has no right children, else left child of the successor,” and that rule lives at generation time, where the framework already has a slot: honesty, the same layer that polices the mint. The architecture is therefore one sided datatype, verified once, policy-free (canonicity, fold exactness, the Join, RA-linearizability), with intent theorems per generation policy: under always-R, read-equivalence to the published RGA (the fragment theorem composed with the compaction theorem); under the Fugue rule, non-interleaving (a subtree is an in-order interval). RGA and Fugue become two disciplines over one kernel rather than two datatypes.

Validation. `whiteboard/litmus/embed_sided.py`: the chain semantics and the bit carrier as lockstep twin models, both run under both policies. Every design prediction survived the first full run:

check	verdict
full gauntlet under the Fugue policy (L1–L25, credential family, subordination CEs, stale forks)	clean, including L19 (out = $\langle 50, 30, 10, 61, 41, 21, 1 \rangle$)
same gauntlet under always-R	clean except L19: the
chain semantics $\sim$ bit carrier, both policies	one-sided fan, reproduced lockstep-EXACT (unique decodability + order embedding)
all-R fragment $\sim$ the one-sided embed model	lockstep-EXACT, every scenario
randomized version-DAG PBT, all four models	120 + 300 clean

The Lean mechanization of both layers (the sided chain-lex kernel and the sided conditioned instance, capstone included) has landed kernel-clean; §8 records it.

6 Comparison with Eg-walker

Eg-walker (Gentle & Kleppmann, EuroSys 2025) is the closest system in spirit, and the two designs can be read as the *operational* and *denotational* realizations of one thesis, **pay for concurrency, not for history**:

	Eg-walker	embedded-chain RGA
model	op-based; replicas exchange events	MRDT: three-way merge with an LCA
durable state	full event graph on disk , deletes included, retained while concurrent ops may arrive; document state metadata-free	live nodes only ; no event log anywhere; dead survive as $\Theta(\log \delta)$ bits inside live coordinates
merge mechanism	<i>replay</i> : walk the event graph, rebuild a transient CRDT structure (tombstones reappear transiently), discard at causally-linear points	<i>refold</i> : $O(\text{survivors})$ coordinate recomputation; no replay, no transient structure, no order decisions
sequential-editing cost	$\approx$ zero (plain edits; no metadata)	$\approx$ zero order metadata (~ 4 bits/level for continuation typing; a cursor jump pays log-gap once, at the run head)
concurrency cost	replay time proportional to the concurrent region, at every affected merge	coordinate bits proportional to $\log(\text{ts gap})$, once, in space
ordering semantics	Yjs-variant (FugueMax-adjacent); maximal non-interleaving <i>conjectured</i> , not proved; two-sided (no L19 analogue)	$\equiv$ published RGA (machine lockstep); one-sided: L19 backward-run interleaving inherited; removed by the sided generalization (§5), RGA kept as the all-R fragment
correctness artifact	informal proof (Appendix C) + property testing	L1–L25 battery + DAG PBT + lockstep vs RGA _† ; RA-linearizability kernel-checked in Lean (§8); the design was chosen so the merge proves itself (no order decisions to verify)
op payload	integer origin positions (compact, human-meaningful)	two timestamps + ghost prefix (droppable on the wire; §3)

Four sentences of honest positioning. (i) Eg-walker achieves its clean document state by keeping the entire history and re-deriving order on demand; the embedded-chain RGA achieves it by making order derivation unnecessary: the sort keys are written at birth and never edited, and the coordinates are their encoding (§2), precomputed at mint time and exact forever. (ii) Eg-walker’s ordering target was stronger where L19 is concerned (Fugue-style two-sidedness removes backward interleaving, while the one-sided design inherits RGA’s anomaly by construction, being read-equal to RGA); the sided generalization of §5 adopts exactly that two-sidedness as a parameter, closing the gap while keeping the one-sided datatype as its all-R fragment. (iii) The verification gap runs the other way: Eg-walker’s replay-transform-clear loop is algorithmically sophisticated and only informally proved; here the merge performs no arbitration at all, the metatheory is one-line at the chain layer plus a single faithfulness theorem at the encoding layer, and RA-linearizability is a kernel-checked theorem in a framework that had already verified a dozen-plus replicated datatypes end-to-end (§8). (iv) The cost models part company at cursor jumps: a single-user history that hops back to type at an old anchor is free for Eg-walker (nothing concurrent to replay) but mints log-gap bits here, once per jump, at the run head; on measured single-user editing traces the jump tail is thin (code bits $\approx 1.4\text{--}1.8$ per level; see the companion note of §2). In the MRDT setting there is also a model fit: Eg-walker must reconstruct merge context by replay because the op-based model gives it none; the MRDT model hands the merge its LCA, which is exactly the context replay rebuilds.

7 Related work

The verified sweep of nearby systems lives in `whiteboard/litmus/related-work.md` (per-claim verdicts, primary sources); this section positions the datatype.

Retention taxonomy. What nearby systems keep for the dead: tombstone records (RGA with cemetery under causal stability; WOOT; Yjs’s per-deletion GC structs, kept forever; Fugue, “we cannot remove a deleted element’s node entirely”; Peritext, “the positions must be remembered”); the full history (Eg-walker’s event graph, Chronofold/causal trees where the log *is* the text); consensus-gated compaction (Treedoc’s flatten requires Paxos-commit); or **dead identifiers inside live position keys** (Logoot/LSEQ, Weidner’s position-strings/list-positions, the Fugue id space). The embedded-chain RGA is in the last class *by information content* (dead ancestors’ timestamps persist inside survivors’ sort keys, at live-reachable scope), but with three deltas to its classmates: the retention is *delta-coded at the entropy of the anchor gaps* (a sequential history costs a constant per element where Logoot-family keys grow with allocation policy), the *birth tree kept whole* gives forward non-interleaving and delete-order preservation (position-key designs interleave, the PaPoC’19 anomaly, and Figma-style fractional indexing is unsound under concurrency by its own account), and the semantics is pinned to the best-studied sequence CRDT by a machine-checked lockstep rather than being a new point in anomaly space.

The impossibility backdrop. Attiya et al. (PODC 2016) prove $\Omega(D)$ metadata for push-based protocols meeting even the weak list spec: the published form of “ordering requires remembering the dead.” This design does not escape the bound; it *relocates and minimizes* the memory: $\Theta(\log \delta)$ bits per dead-chain element inside live coordinates (plus the MRDT version store, which the model provides). The sharper, machine-witnessed form from our design sweep is the *conservation law* (§1): the sort key must retain dead ancestors’ timestamps (the verdict information survives every encoding change: stamp, spine, tombstone, or denominator bits), and ties force separate *metadata*

only under timestamp-oblivious minting; the timestamp-faithful mint carries the same bits inside the coordinate. Eight timestamp-oblivious mutation variants each have a named machine-checked countermodel (fold-verdict erasure; repair non-locality); the credential countermodel separates this design from all of them.

Verified datatypes. Isabelle RGA (Gomes et al. OOPSLA’17, op-based convergence), OpSets (list semantics specified by a total order of identified ops: arbitration-by-identity as specification), VeriF_x (SMT, convergence properties), MET (TLA+, list CRDT bugs). In the MRDT line (Kaki et al. OOPSLA’19; Peepul PLDI’22; RA-lin OOPSLA’25, this framework), no verified sequence with sequence-CRDT-grade ordering guarantees exists; our verified rehoming RGA (§1) is RA-linearizable against its own fold yet fails the sequential delete-order spec, since RA-linearizability certifies convergence to the datatype’s *own* semantics and ordering intent must be specified and proved separately. The embedded-chain RGA delivers the missing artifact: RA-linearizability is a kernel-checked Lean theorem (§8); delete-order preservation, display stability and forward non-interleaving are proved at the model layer; and the RGA read-equivalence (the compaction theorem, §8) closes the intent question against the best-studied sequence semantics.

Treedoc’s lesson, inverted. Treedoc re-balances geometry and must pay consensus for it; the principle that emerged from our ledger-based sibling design study (*reprojection without consensus is safe iff geometry has been demoted to a rendering*) is here taken to its fixed point: geometry is never reprojected at all. It is immutable, so it may safely be the only thing.

8 Mechanization status

The verification framework is the RA-linearizability metatheory for MRDTs (OOPSLA’25), mechanized in the Sal Lean development, the same framework in which twelve production MRDTs (the rehoming RGA among them) and a further half-dozen born-conditioned instances (the Peepul mergeable queue included) are verified end-to-end; file paths below are relative to that repository.

The capstone is proved (2026-07-15, kernel-clean on {propext, Classical.choice, Quot.sound}):

$$\text{embed_ra_linearizable3} : \text{EReach } \Gamma \ C \rightarrow \text{IsRALinearizable3 } C$$

i.e. RA-linearizability per version at every honestly reachable configuration, *parametric in the code* Γ : the unary code, the binary delta code, and the flipped Elias- δ code are all proved **OrderedPrefixCode** instances, three encodings on one proof. The factoring is exactly this note’s design–encoding split, which is itself evidence for the split:

1. **Model layer** (Sal/MRDTs/RGA_Embed/, six files, 0 sorry, plus the read-equivalence order core and the sided kernel noted below): the code-parametric kernel; commutations *to state equality* under $\text{rc} = \text{Either}(\text{rc_non_comm})$; **ins_prefix_ghost**; coherence and chain-generation closures; display stability, with step stability (S2, delete-order preservation at its ‘del’ instance) proved *without any axioms* and merge pairwise stability (S4, the adopted contract); unique decodability (**coordOf_inj**); **the chain-lex theorem** (**display_iff_chainBefore** = Corollary 1); non-interleaving as subtree convexity (**subtree_convex**); Theorem 2(i) for all three codes (**binEnc_mono/binEnc_prefixFree**, **dEnc_mono/dEnc_prefixFree**, cross-validated against the Python codes by kernel **decide**); **native_decide** SPOTs where the rehoming RGA’s delete-reorder witness gets the opposite verdict.

2. **Conditioned instance** (`Sal/ConditionedMRDTs/MRDT_Instances/EmbedRGA/`, nine sections, 0 sorry, plus the Elias- δ inheritance corollary `EmbedRGA_EliasDelta.lean`), proved via the *mergeable-queue route*: a generic join lemma (`JoinLemma3At`: the merge exhibits its own linearization witness) plus an honest-reachability metatheorem (`HonestReach`), rather than the rehoming RGA’s 55-file bespoke chain. The route applies exactly when states are canonical per event set, which is Theorem 1(i). Landed: the canonical sorted-list state; strictly-sorted extensionality (`esorted_ext`: sorted lists with the same members are equal, replacing any canonical-list formula); **fold-canonicity** (`e_fold_canon`: well-formed enumerations of one event set fold to the *same* state; Theorem 1(i) mechanized); the honesty core and its bridge to well-formedness; the merge membership characterization (`e_mergeL_mem`, OR-set survival with honesty closing the cross-branch-delete corner); the Join (`e_join_at`: the merge is its own linearization witness); the capstone; the honesty discharge from the framework’s generation discipline (`eHonest_of_applicable`: applicable generation forces the honesty core, so per-op honesty needs no bespoke oracle); and the instance intent theorems: delete-order preservation *is* sublist stability (`eUpdate_del_sublist` is literally `List.filter_sublist`: a delete’s successor state is a sublist of its predecessor, so the surviving display order is untouched; the merge preserves each input’s survivor order on both sides).

The compaction theorem is proved (2026-07-15, kernel-clean): $\text{read} \circ \text{embed} = \text{read} \circ \text{RGA}_\dagger$ on honest executions, as `rga_read_eq_embed_read` (`EmbedRGA_ReadEquiv.lean`): for every well-formed enumeration of an honest event set, *any* enumeration of the tombstoned fold’s visible ids in the published RGA’s own relational order (`visible_lt`) equals the embed fold’s id sequence, with element agreement (`embed_rec_readSeq`). The proof is *against the published definition*: the order core (`Sal/MRDTs/RGA_Embed/RGA_Embed_ReadEquiv.lean`) shows `visible_lt` coincides with chain-lex on the shared birth tree (soundness by induction on derivations, completeness by realizing every chain prefix as an **afters**-reachable ancestor), and en route proves the published relational order total and asymmetric on the birth tree, facts the published side never had. This converts the design into a theorem *about* the published RGA: it admits a strictly-live compaction preserving every read. The lockstep rows of §4 are this theorem’s tested form.

The sequential specification is proved, and the design now holds the canonical seat (2026-07-16). The sequential-specification campaign (its self-contained note: `Sal/ConditionedMRDTs/seqspec-campaign.pdf`) proved `embed_seq_sound`: on every sequentially honest single-replica history the canonical sorted state IS the naive insert-at-position text buffer, record for record, with the adjacency lemma (`chainBefore_snoc_iff`: a fresh insert’s delta beats every existing child’s by sum-telescoping) as the geometric heart; the published $\text{RGA}_\dagger$ inherits the same buffer spec through the compaction theorem (`rga_seq_read_eq_buffer`). The same statement is REFUTED for the rehoming design (`rehoming_seq_refuted`: the §4 L1 row generalized to a kernel-checked refutation on an honest four-op trace), which demoted it repo-wide: the embedded-chain family is now the repository’s canonical sequence RDT, and the canonical fused Peritext was re-based onto THIS kernel (`Peritext_Embed/`: capstone by pure instantiation of the payload-generic instance; `renderIds_del`, deleting a character never re-formats another, the theorem the rehoming kernel provably lacks; and the tier-4 editor theorem `peritextEmbed_seq_sound`). One caveat is recorded as an open question (oq:seqwitness, campaign note §8): per-replica soundness and convergence do not compose into “the merged read is the naive buffer on some ordering of the union”; the dead-anchor corner is where stamp-sorted replay diverges from the fold.

The sided generalization is proved (2026-07-15, kernel-clean), in the same two layers as §5 presents it. The sided chain-lex kernel (`Sal/MRDTs/RGA_Embed/Sided_ChainLex.lean`): the

sided marker theorem `sdisplay_iff_schainBefore` (display comparison of sided coordinates IS the in-order rule), totality of the display relation *without any axioms*, unique decodability (`sidedCoordOf_inj`: the side is recoverable from the first symbol’s band, the delta by prefix-freedom within the band), the all-R *fragment theorem* `schainBefore_liftR` (all-R chains order exactly as one-sided chains: “RGA is the all-R fragment,” mechanized), and in-order-interval subtree convexity (`schain_subtree_convex`, the datatype half of the L19 discharge). The conditioned instance (`MRDT_Instances/SidedRGA/`, nine sections mirroring the one-sided instance with the sided key): fold-canonicity (`s_fold_canon`), the Join by the same mergeable-queue route, and the capstone

`sided_embed_ra_linearizable3 : SReach  $\Gamma$  C  $\rightarrow$  IsRALinearizable3 C`

for every side assignment: sides are payload to convergence, which is the policy/datatype split of §5 as a theorem, with `sHonest_of_applicable` discharging honesty from the generation guard exactly as one-sided. The per-policy intent theorems have since landed, and the policy story is a machine-checked three-act arc (2026-07-16/17). *Act one:* `sFold_liftOp`, the all-R sided fold IS the one-sided fold, unconditionally (the sort keys coincide definitionally), so every one-sided theorem transports; and `sided_fold_subtree_convex`, subtrees display as contiguous blocks in any chain-generated fold, so runs that chain never interleave (`fugue_reachable_runs_no_interleave`). *Act two:* the Weidner–Kleppmann maximal-non-interleaving Definition 4 was formalized over reachable configurations (`SidedRGA_Fugue.lean`) and REFUTED twice for this Fugue policy on reachable traces (the recency tiebreak reverses the paper’s lowest-ID-first; their Figure-7 execution): the paper’s own Fugue-vs-FugueMax separation, landing on this implementation. *Act three:* FugueMax realized as a kernel variant (`SidedMax_ChainLex.lean`): its sibling order depends on the right ORIGIN, not expressible by any re-banding of deltas (the mirror control machine-witnesses this), so R-entries carry the mint-time successor’s key as an order-reversing tag; condition (3) is proved in full (`fuguemax_same_origin_low_first`, the positive twin of the refutation) and full Theorem 9 is reduced to exactly two stated gaps, the display-to-traversal theory (task #88). The sided sequential spec also landed (`sided_seq_sound`: a sentinel-rooted two-sided buffer, R-inserts splice after their anchor, L-inserts immediately before). The tag has a COST: at contested inserts the coordinate embeds the successor’s key, $\Theta(|\text{key}|)$ where plain Fugue pays zero, and whether that retention is a lower bound for tombstone-free maximal non-interleaving is an open question with a Myhill–Nerode shape (task #89): the entropy law of §2 meeting the ordering intent.

Provenance. `whiteboard/litmus/embed_tree.py` (the design, the timestamp-oblivious control, the credential countermodel, the IEEE 754 measurements; `python3 embed_tree.py / fp / code`); `litmus.py`, `pbt.py` (battery and DAG harness); `contest_tree.py` (lockstep harness, CE regressions); `whiteboard/retention-countermodel.pdf` (the retention arc that forced this design); `whiteboard/delta-tree-design.pdf` (the ledger-based sibling design and the invariants I1–I3); `whiteboard/litmus/related-work.md` (verified source list: Roh et al. JPDC’11; Attiya et al. PODC’16; Kleppmann et al. PaPoC’19; Weidner & Kleppmann TPDS’25 (Fugue); Gentle & Kleppmann EuroSys’25 (Eg-walker, arXiv:2409.14252); Pregoça et al. ICDCS’09 (Treedoc); Grishchenko & Patrakeev PaPoC’20 (Chronofold); Litt et al. CSCW’22 (Peritext); Gomes et al. OOPSLA’17; OpSets 2018; Kaki et al. OOPSLA’19; Peepul PLDI’22; RA-lin OOPSLA’25; VeriFx ECOOP’23; Figma/Wallace fractional indexing; Weidner position-strings).